DualBalance Districting:
From Detection to Generation

Steven Hart

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Abstract

Most current work on gerrymandering tries to detect it. We generate maps instead. *DualBalance Districting* is a deterministic algorithm that draws congressional districts as a pure function of state geometry, census population, and seat count N, with no discretionary parameters, no randomness, and no human input at any stage. Every design choice (seed geometry, objective weighting, tie-breaking rules) is pre-committed and fixed before any specific state's data are examined. Each district carries roughly $1/N$ of the state's people *and* roughly $1/N$ of its land, weighted equally in a single objective called the DualBalance Score (DBS). The procedural neutrality is symmetric: the rule that forecloses partisan engineering equally forecloses race-conscious remediation.

Validated on all 41 multi-seat states for which TIGER 2020PL voting tabulation district boundaries are available, DualBalance achieves *Karcher*-compliant population balance (pop_dev_max $\leq$ 0.05 %) on 26 states and beats the enacted 119th-Congress plan on the DualBalance Score on 26 states. Two deterministic baselines — Cascade (Iowa-LSA county aggregation) and BDistricting (capacitated power diagrams) — achieve *Karcher* compliance on zero states. On partisan fairness, we restrict analysis to the 20 states for which precinct-level composite or congressional election returns are available; on this subset DualBalance reduces |EG| relative to the enacted plan on 14 of 20 states (median |EG| 0.085 vs. enacted 0.133). A House-wide projection across the 41 scored states (369 seats), under uniform-swing assumptions from the available precinct-level vote data, gives DualBalance R 190 / D 177 versus enacted R 196 / D 171 (see §3.2 for caveats).

The conventional gerrymandering metrics — compactness, Efficiency Gap, ensemble outlier tests, majority-minority counts — play two roles on enacted plans: describing plan effects and supporting inferences about the line-drawer's intent. The descriptive role survives on a DualBalance map; the intent-attribution role does not, because there is no line-drawer. We report these diagnostics as descriptions of the partition.

1 Introduction

In most U.S. states, congressional district boundaries are drawn by the same legislators whose electoral prospects depend on where those boundaries fall. This structural conflict of interest is not resolved by the Constitution, which mandates only that House districts achieve population balance (Art. I, §2) while leaving every other line-drawing decision to political actors. The consequences are well-documented: gerrymandered boundaries reduce electoral competition, dilute minority voting power, and produce seat allocations that diverge substantially from statewide vote totals [1, 13]. Federal judicial oversight of these practices has progressively narrowed, as detailed in §1.1 below. This paper proposes and validates *DualBalance Districting*: a deterministic algorithm that computes congressional district maps as a pure function of state geometry, census populations, and district count N, with no discretionary parameters, no randomness, and no human intervention at any stage. Each district is designed to carry approximately $1/N$ of the state's population and approximately $1/N$ of its land area, weighted equally in a single objective we call the DualBalance Score. A deterministic algorithm with pre-committed, publicly documented rules provides, by construction, a mathematical barrier against ad hoc map manipulation: every design choice is fixed before any specific state's data are examined, so no choice can be made after the fact to favor a party or candidate.

1.1 Redistricting Without a Neutral Standard

The federal courts have largely withdrawn from gerrymandering review. In *Rucho v. Common Cause* [19], the Supreme Court held 5–4 that partisan-gerrymandering claims present nonjusticiable political questions; the Court conceded the practice is "incompatible with democratic principles" but found no manageable judicial standard. After *Rucho*, the only federal channel for gerrymandering review is racial. That channel has since narrowed. Section 2 of the Voting Rights Act, as construed in *Thornburg v. Gingles* [17], requires majority-minority districts where the three Gingles preconditions are met. The Court reaffirmed this in *Allen v. Milligan* [20]. One year later, *Alexander v. South Carolina Conf. of the NAACP* [22] raised plaintiffs' evidentiary burden; *Louisiana v. Callais* [23] held in April 2026 that even a plan drawn to comply with a prior Section 2 order may violate the Equal Protection Clause if Section 2 did not in fact compel a race-based remedy. Together, *Alexander* and *Callais* leave race-conscious line-drawing in a narrower window of constitutional safety than at any point since the VRA's adoption.

State constitutional review under *Moore v. Harper* [21] remains available but is uneven and politically contingent: roughly ten state supreme courts have recognized state-constitutional partisan-gerrymandering claims [9]; the rest have not. Several states have already redrawn maps mid-decade in response to electoral results rather than census revision, shifting redistricting from a once-per-decade event into a recurring one. Reducing this discretionary intervention requires a procedure whose output is determined entirely by census data, with no human choices at the line-drawing stage.

1.2 Population and Geography as Dual Representational Axes

The Constitution already encodes two theories of representation: the House apportions seats by population (Art. I, §2), the Senate by sovereign state regardless of population (Art. I, §3). Madison's defense in *Federalist* Nos. 54–58 [6] frames the bicameral structure as a principled refusal to collapse representation onto a single dimension. Within the House, the existing framework uses only one extensive quantity, population, to define what a district should be. Geographic area is a second extensive quantity: it is well-defined, measurable from the same census geometry, and orthogonal to population density. Balancing area alongside population discourages concentration of districts exclusively within either dense metropolitan cores or sparse rural regions.

The Electoral College illustrates how the founders embedded both principles within a single institution. Each state's electoral vote total equals its House seats plus its two Senate seats (Art. II, §1), blending a population-proportional component with a geographic floor. The analogy operates at the level of constitutional design philosophy, not mechanical equivalence: the Electoral College combines the two principles *across* states, allocating a fixed geographic bonus between them, while DualBalance applies them *within* a single state's districts. The design logic is the same regardless, a constitutional order that declines to collapse representation onto a single extensive quantity.

This is a mathematical challenge, not a free choice. A typical U.S. state varies by two to three orders of magnitude between its densest census tract and its sparsest rural block. Equal population, equal area, contiguity, and shape compactness cannot all be satisfied simultaneously in general; some trade-off is unavoidable. The question is which trade-off to make, how transparently to make it, and whether the resulting procedure can be defended as impartial.

1.3 Existing Metrics Are Forensic, Not Generative

Quantitative tests for gerrymandering include shape-compactness measures (Polsby-Popper [10]; Reock [11]), partisan-asymmetry statistics [28, 29], the Efficiency Gap [13], and ensemble outlier methods [3, 4]. We report Polsby-Popper, Reock, and the Efficiency Gap (EG) as comparative benchmarks in §3. EG, defined formally in §2, measures the difference between the two parties' wasted-vote totals as a fraction of statewide turnout; it has received the widest adoption in federal redistricting litigation, serving as the central metric in *Whitford v. Gill*, with $|\text{EG}| > 0.07$ proposed as a threshold for a plausible partisan-gerrymandering challenge [13]. All of these metrics are *forensic* instruments: they take an enacted map as input and ask whether its observed properties (shape, vote efficiency, partisan asymmetry, the location of its seat–vote curve) are consistent with maps that some reference process (random redistricting, ensemble sampling under legal constraints) would have produced absent partisan intent. They compare the enacted plan against a counterfactual distribution to infer the line-drawer's motive, and were built for an adversarial context in which the question is "did somebody do something here that they should not have done?"

A generative metric is a different object. It states, directly, what a district *should* look like, and is used as the objective of the line-drawing procedure rather than as evidence about the procedure. Population balance is the only generative metric U.S. redistricting law currently uses: *Wesberry v. Sanders* [15] and *Reynolds v. Sims* [14] require that each district hold roughly $1/N$ of the relevant population. Every other criterion (compactness, communities of interest, partisan fairness) enters the law as a forensic instrument or a discretionary guideline.

The DualBalance Score (DBS; equation 1 in §2) is a *generative* criterion: it states what a district should be, one that carries roughly $1/N$ of the people and roughly $1/N$ of the land, and serves as the objective the algorithm pursues rather than a test applied after the fact.

1.4 The Iowa Model as a Reference Point

Iowa's Legislative Services Agency [5] has operated under a written, publicly documented redistricting procedure since 1980 – one of the few U.S. examples of a formalized, procedural approach that has functioned without major legal challenge across multiple redistricting cycles. Its longevity and transparency make it a natural reference point for proposals to replace discretionary redistricting with an explicit algorithm. The open question is whether a procedure designed for Iowa's geographic and demographic conditions constitutes a legally defensible, impartial template for states with substantially different population distributions and geographic structure. We re-implement it as a structured baseline (referred to here as *Cascade* to avoid confusion with the agency) and use it as a point of comparison throughout. Cascade captures the geographic-prioritization logic of the Iowa formula (county integrity first, then population balance, then compactness) but omits the institutional framework (the bipartisan advisory commission, multiple plan submissions, and legislative up-or-down vote) that is integral to Iowa's actual redistricting process. The comparison therefore isolates the algorithmic contribution of county-preservation, not the process surrounding it.

1.5 Contribution

We propose *DualBalance Districting*: a deterministic multi-resolution pipeline whose output is a pure function of $(\text{state geometry}, \text{census populations}, N)$, formalized in §2.

Pipeline. The algorithm runs in three stages: radial seed placement around the population-weighted centroid, followed by a two-phase greedy local search at VTD scale that closes per-district population deviation toward the *Karcher* threshold [16], followed by the same search re-run at census-block scale where finer granularity allows the DualBalance Score to make its largest gains. The full formalization is in §2.

Empirical validation. We evaluate the full pipeline on all 41 multi-seat states for which TIGER 2020PL VTD boundaries are available (California, Hawaii, and Oregon are excluded for lack of VTD data). DualBalance achieves *Karcher*-compliant population balance (pop_dev_max $\leq 0.05\,\%$) on 26 of 41 states and exceeds the enacted 119th-Congress plan on the DualBalance Score on 26 of 41 states. On partisan fairness, DualBalance reduces |EG| relative to the enacted plan on 14 of the 20 states with composite or congressional election data (median |EG| 0.085 vs. enacted 0.133). On minority-majority district counts, DualBalance produces at least as many as the enacted plan on most states and more on several high-minority states, including TX (22 vs. 19 enacted) and NC (2 vs. 1 enacted).

Claims. The procedure contains no explicit partisan, racial, or incumbent-aware optimization criteria: its inputs are census geometry and population totals, and its output is a pure function of those inputs. Every design choice, including seed geometry, objective weighting, tolerance, and tie-breaking, is pre-committed and fixed before any specific state is examined. There are no in-session discretionary parameters that a line-drawer could adjust to affect electoral or demographic outcomes. Adopting a generative criterion as the design objective, rather than a forensic instrument, is a natural response to a setting in which no line-drawer is present to investigate. Scope limits and further qualifications appear in §4.

Roadmap. Section 2 formalizes the algorithm. Section 3 reports the six-state empirical validation. Section 4 discusses limits, trade-offs, and legal exposure.

2 Methods

2.1 Data and inputs

Atomic units are U.S. Census Bureau *voting tabulation districts* (VTDs) and *census tabulation blocks* from the 2020 decennial release [26], obtained via TIGER/Line shapefiles [25]. Boundaries are used as-is; no simplification or smoothing is applied. Population counts are the total-population figures from PL 94-171. The enacted congressional plan used for comparison is the 119th-Congress plan (effective January 2025), joined to VTDs and blocks via a spatial representative-point join against the TIGER 2024 `cd119` layer [27].

Note on enacted-plan population deviations. Enacted U.S. congressional plans are litigated to *Karcher* compliance at the census-block level. The enacted-plan deviations reported in this paper reflect the VTD-layer spatial join: a VTD that straddles a district boundary is assigned in full to one district, introducing apparent deviation where the legislative record shows none. DualBalance's reported deviations come from its own block-refined output and are directly comparable to the *Karcher* threshold. Counts of enacted-plan Karcher compliance (Tables 1–2) should be read as a conservative lower bound on enacted compliance, not as legally cognizable deviations: the true enacted deviations are near zero on block-accurate data.

Reproducibility. The `dualbalance` Python package, per-state configuration files (`configs/`), and the data pipeline (`scripts/prep_state_units.py`) reproduce all tables and figures from publicly available TIGER and Census source files. Algorithm outputs are byte-identical across repeated runs on the same input data; the `votes_source` field in each state's units file records which election data source was used.

2.2 Problem statement

Given a state S partitioned into atomic units $U = \{u_1, \ldots, u_M\}$ (census tabulation blocks, block groups, or voting tabulation districts (VTDs), in increasing order of size) and a target district count N, DualBalance Districting produces a deterministic assignment $\pi : U \to \{1, \ldots, N\}$ such that each district $D_i = \pi^{-1}(i)$ is contiguous, non-empty, and as close as the geometry allows to representing both $1/N$ of the state's people and $1/N$ of its geography. Let $P = \sum_u \mathrm{pop}(u)$ and $A = \sum_u \mathrm{area}(u)$ with per-district targets $P^* = P/N$ and $A^* = A/N$.

The full pipeline runs in three stages: a radial-seed stage, *DualBalance* (§§2.3–2.5), which provides the deterministic starting configuration; a VTD-scale tightening stage (§2.7) which drives $\max_i |\delta_i|/P^*$ toward the user-supplied tolerance τ via a hybrid greedy local search with bounded-chain escape; and a block-scale refinement stage (§2.8) which re-initializes at finer granularity to recover area balance within the achieved pop-deviation envelope. Each stage is a pure function of its inputs.

2.3 Radial seed placement

Compute the population-weighted centroid

$$c = (c_x, c_y) = \left(\frac{\sum_u p_u \, x_u}{\sum_u p_u}, \; \frac{\sum_u p_u \, y_u}{\sum_u p_u} \right),$$

where x_u, y_u are the centroid coordinates of unit u in an equal-area projection and p_u its population. Let diag denote the bounding-box diagonal of the units and set the seed radius $r = 0.001 \cdot \mathrm{diag}$. For $d = 0, 1, \ldots, N-1$ place seed s_d at

$$s_d = \big(c_x + r\cos(2\pi d/N),\; c_y + r\sin(2\pi d/N)\big).$$

Seed 0 sits due east of the centroid; seeds advance counter-clockwise by equal angular steps. The choice $r = 0.001 \cdot \mathrm{diag}$ is small enough that the Voronoi cells generated by the seeds degenerate to near-perfect radial slices through c, but large enough to keep the seed positions numerically distinct.

These angular parameters, seed 0 due east and counter-clockwise advance, are pre-committed design choices fixed before any state-specific data are examined. Different anchor angles $\theta \in [0, 2\pi/N)$ produce different partitions with different partisan and demographic properties, because the resulting radial slices land differently on the state's population geography. Rotation sensitivity, i.e., the distribution of DBS, |EG|, and seat outcomes across the full rotation range, is therefore an important robustness check identified as future work in §4.6. The due-east anchor was selected as the natural zero of the standard angular convention; an adopting body would need to pre-specify a rotation rule (or adopt this one) as part of any legislative implementation.

The radial configuration is what carries the dual-balance property: each slice spans both the dense (near-c) and sparse (boundary-side) territory of the state, so the population it inherits is bounded by the cap (§2.4) while the area it inherits is driven toward A^* by the slicing geometry.

2.4 Capacitated first-fit assignment

Let $d(u,i) = \|x_u - s_i\|$ be the Euclidean distance from unit u to seed i. Initialize remaining capacities $\rho_i = P^*$ for $i = 1, \ldots, N$. Sort all (u,i) pairs by ascending normalized distance $d(u,i)/\text{diag}$ and walk the sorted list in order:

$$
\begin{array}{l}
\text{for each } (u,i) \text{ in ascending } d(u,i)/\text{diag}: \\
\quad \text{if } u \text{ already assigned: continue} \\
\quad \text{if } \rho_i \geq p_u\text{: assign } u \text{ to } i; \quad \rho_i \leftarrow \rho_i - p_u \\
\quad \text{else: skip}
\end{array}
$$

Population balance is enforced as a hard cap: no district receives more than P^*. Any unit not placed by the end of the walk (a rare integer-rounding edge case) is assigned to the district with the largest remaining capacity; `argmax` resolves ties to the smallest district id.

Ties in normalized distance break by ascending (unit_id, district_id). There is no Lloyd recentering, no iteration count, no convergence test: the radial seeds do not drift, so a single assignment pass suffices.

2.5 Contiguity repair

After §2.4 every unit belongs to exactly one district, but a district may consist of more than one connected component (rare on convex states; more common on those with peninsulas, islands, or rural enclaves). For each such district, the largest connected component by total population is retained; units in the smaller components are reassigned to neighboring districts one at a time, in the lowest-cost direction. Cost is

$$c(u,j) = \frac{d(x_u, s_j)}{\text{diag}} + \frac{|P(D_j) + p_u - P^*|}{P^*} + \frac{|A(D_j) + a_u - A^*|}{A^*},$$

combining a normalized distance term with normalized population and area deviation penalties for the receiving district. Ties break in cascade $(c, \text{pop\_pen}, \text{area\_pen}, \text{dist}, \text{district\_id})$ ascending. The repair sweep iterates until no district has more than one connected component, capped at ten sweeps; in practice it converges in zero or one sweep on real census geometries.

Algorithm 1 summarizes the full core pipeline.

2.6 Scoring

Define per-district relative deviations $\text{pop\_dev}_i = |P(D_i) - P^*|/P^*$ and $\text{area\_dev}_i = |A(D_i) - A^*|/A^*$ and let $\overline{\text{pop\_dev}}$, $\overline{\text{area\_dev}}$ denote their means over $i = 1, \ldots, N$. The DualBalance Score is

$$\text{DBS} = \frac{1}{1 + \frac{1}{2}\,\overline{\text{pop\_dev}} + \frac{1}{2}\,\overline{\text{area\_dev}}}. \tag{1}$$

The 0.5/0.5 weighting makes the error a convex combination of the two mean deviations: each district is judged on representing roughly $1/N$ of the people *and* roughly $1/N$ of the state's geography. The score reaches 1.0 for a perfectly balanced plan and approaches 0 as deviations grow

Algorithm 1 DualBalance Districting

Require: Atomic units $\mathcal{U} = \{u_1, \ldots, u_M\}$ with populations $\{p_u\}$, centroids $\{x_u\}$, and areas $\{a_u\}$; district count N

Ensure: Assignment $\pi : \mathcal{U} \to \{1, \ldots, N\}$; every district contiguous and non-empty

$c \leftarrow \big(\sum_u p_u\, x_u\big) / \big(\sum_u p_u\big)$ ▷ population-weighted centroid
$P^* \leftarrow \big(\sum_u p_u\big)/N;\ A^* \leftarrow \big(\sum_u a_u\big)/N$ ▷ per-district targets
$\Delta \leftarrow \mathrm{diag}\big(\mathrm{bbox}(\mathcal{U})\big);\ r \leftarrow 0.001\,\Delta$ ▷ bounding-box diagonal; seed radius

Phase 1 Radial seed placement

for $d = 0, \ldots, N-1$ **do**
  $s_d \leftarrow c + r\,\big(\cos(2\pi d/N),\ \sin(2\pi d/N)\big)$
end for

Phase 2 Capacitated first-fit assignment

$\rho_i \leftarrow P^*$ for all i; $\pi(u) \leftarrow \varnothing$ for all u
Sort pairs $(u,\, i)$ by $\|x_u - s_i\|/\Delta$ ascending ▷ ties: unit id, then district id
for each pair $(u,\, i)$ in sorted order **do**
  if $\pi(u) = \varnothing$ **and** $\rho_i \geq p_u$ **then**
    $\pi(u) \leftarrow i;\ \rho_i \leftarrow \rho_i - p_u$
  end if
end for
Assign each unassigned u to $\arg\max_i\ \rho_i$ ▷ ties: lowest district id

Phase 3 Contiguity repair

while $\exists\, i$ with more than one connected component **do**
  Let C be the smallest component of district i
  for each $u \in C$, each district $j \neq i$ adjacent to u **do**
    $\mathrm{cost}(u,\, j) \leftarrow \dfrac{\|x_u - s_j\|}{\Delta} + \dfrac{|P(D_j) + p_u - P^*|}{P^*} + \dfrac{|A(D_j) + a_u - A^*|}{A^*}$
  end for
  $\pi(u) \leftarrow \arg\min_j\ \mathrm{cost}(u, j)$ for each $u \in C$ ▷ ties: pop pen, area pen, dist, district id
end while

without bound. The equal weighting is an explicit design choice, pre-committed before any state-specific data are examined. The pre-commitment is the critical property: once a weighting is fixed publicly, neither the designer nor any adopting body can adjust it to favor a particular geographic outcome. The symmetric 0.5/0.5 choice reflects an impartiality principle: a designer who did not know whether their state's geography would favor population-concentrated or area-concentrated outcomes would choose the weighting that gives neither axis an a priori advantage. A weight of 1.0 on population and 0.0 on area recovers population-only optimization; an asymmetric choice between these poles requires a principled justification for why one representational axis should subordinate the other within the same chamber. In the absence of such a justification, 0.5/0.5 is the principled default pending democratic deliberation over the weighting itself. Secondary metrics (Polsby-Popper compactness [10] and Reock [11]) are computed alongside but not optimized against; the Phase 2 optimizer (§2.7) hill-climbs DBS, not the compactness metrics. Radial slices have lower compactness than blob-Voronoi or hand-drawn districts by construction; this is a deliberate trade in service of the dual-balance objective.

DualBalance does not directly minimize (1); it minimizes population-capacitated geographic assignment cost under radial seeding. The empirical consequences are reported in §3.

For partisan-fairness comparison we report the Efficiency Gap (EG) [13]. Let $\text{wasted}_P(d)$ denote the wasted votes for party P in district d: all votes cast for a losing candidate, plus votes above the bare majority threshold for a winning candidate. Then

$$\text{EG} = \frac{\sum_d \text{wasted}_R(d) - \sum_d \text{wasted}_D(d)}{\sum_d \text{total}(d)}, \tag{2}$$

where positive values indicate Republican advantage and negative values indicate Democratic advantage. EG is reported as a diagnostic and does not enter the generator; all partisan, racial, and demographic variables are excluded from the generator's inputs.

Vote inputs for EG are drawn from the best available precinct-level source per state, in priority order: (1) DRA's 2016–22 composite election (`E_16-22_COMP`), which blends presidential, Senate, and gubernatorial returns across three cycles to mitigate incumbency effects and uncontested-race artifacts [2]; (2) 2022 U.S. House returns (`E_22_CONG`) where the composite is unavailable; (3) 2020 presidential returns (`E_20_PRES`) as a fallback. The composite is available for 18 of 41 states, the 2022 House-only for 2 states, and the 2020 presidential for the remaining 21. The `votes_source` field in each state's units file records which source was used; Table 1 marks rows where the fallback presidential source applies ($\star$).

2.7 $\mathbf{L^1}$ population tightening and DBS hill-climb

The radial pipeline produces per-district pop deviation in the 5–15% range on real census geometry, well above the ˜0.5% threshold required by *Reynolds v. Sims* for U.S. congressional districts [14]. A deterministic two-phase local search of boundary-unit moves (activated via `--tighten-pop`) closes this gap; all results reported in this paper use this pass. Let $\delta_i = P(D_i) - P^*$ denote the signed population deviation of district D_i, let T denote the user-supplied tolerance (default 0.005), and call a move *safe* if the source unit is not an articulation point of its district's induced subgraph (so that removing it preserves contiguity).

Phase 1: population tightening (hybrid $\mathbf{L^1}$ + max-norm). At each step, scan every boundary unit and every neighboring district and compute both the L^1 change

$$\Delta_{L^1}(u, d_{\text{dest}}) = |\delta_{d_{\text{src}}} - p_u| - |\delta_{d_{\text{src}}}| + |\delta_{d_{\text{dest}}} + p_u| - |\delta_{d_{\text{dest}}}|$$

and the max-norm effect: a move *strictly reduces* the global maximum iff $|\delta_{d_{\mathrm{src}}} - p_u| < \max_i |\delta_i|$, $|\delta_{d_{\mathrm{dest}}} + p_u| < \max_i |\delta_i|$, and at least one of $\delta_{d_{\mathrm{src}}}, \delta_{d_{\mathrm{dest}}}$ is itself at the current maximum. Each scanned candidate is labeled with a priority key: priority 0 if it strictly reduces the max, priority 1 if it only reduces L^1. Candidates are ranked lexicographically by $(\mathrm{priority}, \Delta_{L^1})$, and the safe top-ranked candidate is accepted.

This hybrid is essential. Pure L^1-greedy leaves the worst district untouched when it sits between two other over-target districts: draining it requires L^1-neutral moves that the L^1 criterion rejects. Pure max-norm gets stuck symmetrically: when two districts are tied at the max, no single move strictly reduces the global maximum (draining one leaves the other at the same value). The hybrid takes any move that helps either norm, with max-reducing moves preferred so the worst district is targeted whenever possible.

When neither 1-opt criterion finds an improving move but $\max_i |\delta_i|/P^* > T$, invoke a bounded *chain escape*: search for an augmenting transport on the district-adjacency graph of the form

$$u_0 : D_0 \to D_1,\ u_1 : D_1 \to D_2,\ \ldots,\ u_{k-1} : D_{k-1} \to D_k,$$

of length $k = 2$ then $k = 3$, in which each u_j is a boundary unit between D_j and D_{j+1}. The chain is accepted if it strictly cuts the global maximum, or if it holds the maximum flat while strictly reducing the L^1 sum (which is what happens when several districts are tied at the max: each chain drains one of them, the global maximum stays at the same value until the last tied district is drained, but the L^1 sum drops monotonically). The unit-level search inside each district triple sorts the arc lists $B_{i\to j}$ by population value; matching pairs (u, v) are found via binary search around a target population value rather than by Cartesian enumeration, bounding the per-triple cost by $O(|B_{i\to j}| \log |B_{j\to k}|)$. Application is in *reverse* order (u_{k-1} first, then u_{k-2}, etc.) so the intermediate districts never temporarily overload during the chain. This is the deterministic analogue of an ejection chain. Phase 1 terminates only when neither 1-opt nor bounded-chain escape finds an improving sequence.

The L^1 objective is essential to the radial geometry. Seed placement puts the most over-target and most under-target districts on opposite sides of the population centroid, so no single adjacent-slice swap reduces the L$^\infty$ maximum, but many such swaps reduce the sum. A pure max-norm criterion stalls on this geometry; the hybrid with chain escape continues to make progress where the L$^\infty$-only criterion would halt.

Phase 2: DBS hill-climb. Once Phase 1 has converged, at each step pick the safe boundary move that maximally improves DBS (equation 1), subject to $\max_i |\delta_i|/P^*$ not exceeding the value Phase 1 reached (equivalently, never crossing back above T). Phase 2 runs until no improving safe move exists. Whereas the core DualBalance pass does not directly minimize equation 1, this opt-in Phase 2 does, but only on the post-Phase-1 plan and only within the Reynolds-compliant feasible region.

Determinism and engineering. Both phases are pure functions of (units, N, T); the full pipeline including the rotation anchor is a pure function of (units, N, θ, T) where $\theta = 0$ in all results reported here. The only source of dependency beyond the core pipeline's (units, N, θ) is the explicit tolerance T. The optimizer maintains a per-district articulation-point cache via Tarjan's algorithm on CSR adjacency arrays [24], reducing the per-candidate contiguity check from $O(V + E)$ to $O(1)$; an incrementally tracked boundary-unit set restricts each scan to units that have a different-district neighbor. At VTD scale the speedup is modest; at block scale (tens of thousands of units per district) it is the difference between hours and minutes.

The pass is off by default and gated by an explicit flag. Pure radial remains the principled algorithm; the two-phase optimizer is an opt-in concession to legal compliance, with Phase 2 providing the further courtesy of directly hill-climbing the project's stated objective once the Reynolds constraint is satisfied.

2.8 Multi-resolution refinement: VTD to Census block

The hybrid Phase 1 + chain escape closes most of the Reynolds gap at VTD scale, but not all of it. Modern U.S. congressional plans must clear the much tighter *Karcher v. Daggett* [16] practical threshold of $\sim 0.05\%$ total deviation, and at VTD granularity the residual gap is structural rather than algorithmic. The smallest single-unit move has the size of the smallest VTD's population. On real states the median VTD population is on the order of 10^3, while the Karcher budget on a $\sim$ 800k ideal district is ~ 400 people. Most candidate moves overshoot the budget on at least one endpoint, and Phase 2 starves: the running-max constraint admits few moves, and the algorithm settles above Karcher with most of the residual area-balance gain unrealized.

The fix is to refine at finer granularity. We re-run the optimizer on Census tabulation blocks (median population ~ 20 rather than $\sim 10^3$), initializing from the VTD-Karcher plan rather than re-seeding from scratch. The pipeline is:

1. Compute the VTD-level plan $\pi_{\text{VTD}} : V_{\text{VTD}} \to \{1, \ldots, N\}$ via DualBalance + Phase 1 + Phase 2 at VTD scale with tolerance T.
2. Load the block-level units V_{block} and project both layers to the same equal-area CRS.
3. For each block $u \in V_{\text{block}}$, find the VTD polygon $v(u) \in V_{\text{VTD}}$ containing its representative point, and set $\pi_0(u) := \pi_{\text{VTD}}(v(u))$. Blocks whose representative point lies outside every VTD polygon (rare, only on coastal/water boundary cases) fall back to the nearest VTD by centroid distance. The spatial join is deterministic given a fixed tie-breaking order on unit_id.
4. Run Phase 1 + Phase 2 again at block scale starting from π_0, with the same tolerance T.

Because VTDs are unions of whole blocks, π_0 inherits the VTD-level plan's pop totals and contiguity exactly: the block-level plan is identical to the VTD-level plan re-described at finer granularity. Phase 1 at block scale therefore has nothing to do on states where the VTD-level pass reached T. Phase 2 has substantial freedom: block populations are two orders of magnitude smaller than VTD populations, so the running-max envelope around T admits many candidate moves and the DBS hill-climb runs until saturation.

Because DualBalance-style seeding at block scale starts from a much worse initial configuration than the VTD seed (the population-weighted centroid is the same but the much higher density of degenerate articulation-point boundaries traps the optimizer), the refinement strategy is the path that works in practice: VTD as a coarse solver, blocks for the precise area-balance recovery. Contiguity checks within the optimizer use a per-district articulation-point cache (Tarjan's algorithm [24] on CSR adjacency arrays), reducing each query from $\mathcal{O}(|V_d| + |E_d|)$ to $\mathcal{O}(1)$ amortized; this is what makes the block-scale stage tractable on commodity hardware.

3 Results

We evaluated DualBalance against the enacted 119th-Congress plan on all 41 multi-seat states for which TIGER 2020PL VTD boundaries are available. California, Hawaii, and Oregon did not submit VTD data to the Census Bureau and are excluded throughout (marked † in figures and tables). The data pipeline is automated via `scripts/prep_state_units.py`; for election data it selects the best available precinct-level source per state (see §2.7). Two additional deterministic algorithms serve as baselines: *Cascade* (`src/dualbalance/cascade.py`), an Iowa-LSA-flavored construction that aggregates VTDs to counties; and *BDistricting* [7], Brian Olson's published 50-state maps ingested via Census 2020 Block Assignment Files (`scripts/prep_bdistricting.py`). All algorithm outputs are byte-identical across repeated runs.

Results are reported on two tiers. **Non-partisan metrics** (DualBalance Score, population deviation, compactness, majority-minority districts) use all 41 states. **Partisan-fairness metrics** (Efficiency Gap) are restricted to the 20 states for which DRA composite or 2022 congressional election returns are available at VTD scale; the remaining 21 states use 2020 presidential returns as a proxy and are marked $\star$ in Table 1.

Three cross-state findings organize the presentation.

Partisan asymmetry shrinks under every deterministic generator. Among the 20 states with composite or congressional election data, DualBalance reduces |EG| relative to the enacted plan on 14 of 20 states, with a cross-state median |EG| of 0.085 versus 0.133 for the enacted plans (Figure 1). All three deterministic algorithms produce a lower median |EG| than the enacted plans on this subset. The structural explanation is in §4.2: a generator that reads no political data cannot reproduce the systematic wasted-vote asymmetry that characterizes enacted gerrymanders.

Cascade wins on DBS but is legally non-viable on most states. Cascade scores well on the DualBalance objective because county aggregation naturally produces units of similar area, but the county-integrity constraint yields population deviations far above the *Karcher* threshold on any state with a dominant metropolitan county (Figure 2, Panel B). DualBalance achieves *Karcher* compliance on 26 of 41 states; BDistricting and Cascade achieve it on zero states (Table 2). Cascade clears *Karcher* nowhere because the county-integrity constraint produces large deviations whenever a county exceeds the per-district cap. BDistricting clears it nowhere because Lloyd iteration converges to centroidal districts that do not enforce strict population equality. A plan that wins on DBS but violates *Reynolds v. Sims* cannot legally be enacted.

Population compliance tiers. The 15 states that fall short of *Karcher* divide into two qualitatively distinct groups. Twelve sit in an algorithmic-convergence gap between the *Karcher* threshold and roughly 1 %: the block-scale refinement stage (§2.8) is the designed path for closing this gap. Three states, Florida (44.2 %), New York (13.3 %), and West Virginia (10.4 %), are geometric failure cases where single-center radial seeding cannot approach the legal threshold regardless of refinement: Florida and New York have polycentric population distributions that a single centroid cannot span, and West Virginia has a narrow panhandle geometry that forces the optimizer into a structural dead end. These states are included in aggregate statistics for completeness but are non-viable under the current pipeline; multi-center seeding (§4.6) is the structural fix.

Different deterministic algorithms, different trade-offs. DualBalance holds pop_dev_max below 1 % on nearly all states and recovers area balance through radial mixing of dense and sparse territory, at the cost of lower Polsby-Popper compactness (Figure 2, Panels A and D). BDistricting maximizes compactness and population balance but accepts high area imbalance because Lloyd recentering explicitly minimizes within-district geographic spread. Cascade maximizes county integrity and area balance but fails the population-balance legal threshold on most states.

Table 1: Per-state DualBalance Score and maximum population deviation for all 41 states with TIGER 2020PL VTD data. **Bold** marks the highest DBS per row. ‡: Cascade pop_dev_max > 0.5 % (*Reynolds* non-compliant). ⋆: Efficiency Gap for this state computed from 2020 presidential returns (composite/congressional data unavailable); EG not shown.

State	N	DualBalance Score				pop_dev_max	
		DB	Cascade	BDist	Enacted	DB	Cascade
Alabama	7	0.828	0.741‡	0.816	**0.880**	0.04%	*93.73%*‡
Arkansas⋆	4	**0.890**	0.799‡	0.755	0.759	0.05%	*55.50%*‡
Arizona	9	**0.678**	0.637‡	0.628	0.651	0.07%	*50.08%*‡
Colorado⋆	8	0.624	**0.791**‡	0.634	0.648	0.05%	*98.50%*‡
Connecticut⋆	5	0.799	—	0.784	**0.806**	0.14%	—
Florida	28	**0.734**	0.675‡	0.689	0.716	44.20%	*88.06%*‡
Georgia	14	0.700	0.703‡	0.715	**0.717**	0.13%	*99.09%*‡
Iowa⋆	4	0.858	**0.899**	0.788	0.883	0.05%	0.29%
Idaho⋆	2	0.866	0.638‡	0.804	**0.977**	0.04%	*48.77%*‡
Illinois	17	**0.664**	0.642‡	0.636	0.645	0.05%	*74.52%*‡
Indiana	9	**0.807**	0.786‡	0.785	0.801	0.04%	*64.45%*‡
Kansas	4	0.708	0.679‡	0.695	**0.737**	0.05%	*82.83%*‡
Kentucky⋆	6	**0.806**	0.789‡	0.760	0.784	0.03%	*42.54%*‡
Louisiana	6	**0.838**	0.822‡	0.817	0.836	0.05%	*54.67%*‡
Massachusetts⋆	9	**0.790**	0.718‡	0.716	0.725	0.12%	*41.56%*‡
Maryland⋆	8	0.697	**0.762**‡	0.664	0.688	0.05%	*71.88%*‡
Maine⋆	2	**0.932**	0.688‡	0.721	0.731	0.10%	*60.48%*‡
Michigan⋆	13	0.646	**0.677**‡	0.630	0.636	0.04%	*84.22%*‡
Minnesota	8	0.661	**0.696**‡	0.649	0.639	0.06%	*76.14%*‡
Missouri⋆	8	**0.771**	0.734‡	0.724	0.716	0.04%	*92.35%*‡
Mississippi	4	0.884	0.822‡	**0.910**	0.884	0.04%	*31.40%*‡
Montana	2	**0.861**	0.759‡	0.820	0.819	0.02%	*15.57%*‡
North Carolina	14	0.753	**0.811**‡	0.760	0.769	0.11%	*10.27%*‡
Nebraska⋆	3	**0.776**	0.616‡	0.646	0.634	0.00%	*89.37%*‡
New Hampshire⋆	2	0.747	0.763‡	0.731	**0.803**	0.01%	*55.95%*‡
New Jersey⋆	12	**0.732**	0.643‡	0.693	0.725	0.04%	*67.97%*‡
New Mexico	3	0.762	0.654‡	0.754	**0.840**	0.03%	*92.55%*‡
Nevada	4	**0.711**	0.674‡	0.671	0.678	0.05%	*6.27%*‡
New York	26	**0.630**	0.586‡	0.626	0.611	13.25%	*74.03%*‡
Ohio	15	**0.766**	0.754‡	0.716	0.744	0.05%	*63.52%*‡
Oklahoma⋆	5	**0.748**	0.726‡	0.745	0.717	0.05%	*84.51%*‡
Pennsylvania⋆	17	**0.728**	0.681‡	0.687	0.680	0.05%	*74.42%*‡
Rhode Island⋆	2	**0.958**	0.849‡	0.712	0.853	0.01%	*10.66%*‡
South Carolina	7	**0.902**	0.768‡	0.819	0.848	0.04%	*71.80%*‡
Tennessee⋆	9	0.815	0.759‡	0.761	**0.816**	0.55%	*92.20%*‡

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State	N	DualBalance Score				pop_dev_max	
		DB	Cascade	BDist	Enacted	DB	Cascade
Texas	38	0.692	**0.694**‡	0.635	0.666	0.90%	*24.58%*‡
Utah⋆	4	0.742	0.694‡	0.658	**0.761**	0.05%	*79.77%*‡
Virginia⋆	11	0.741	0.741‡	0.704	**0.746**	0.19%	*61.26%*‡
Washington	10	0.697	0.702‡	0.698	**0.716**	0.02%	*54.03%*‡
Wisconsin	8	0.695	**0.802**	0.742	0.741	0.04%	0.50%
West Virginia⋆	2	**0.904**	0.827‡	0.867	0.880	10.44%	*6.48%*‡

Table 2: Aggregate algorithm comparison. **Bold** marks the best value per row. Non-partisan rows use all 41 available states unless noted. ⋆ Efficiency Gap rows restricted to 20 states with composite or congressional election data. † Enacted pop_dev_max and *Karcher* counts reflect the VTD-layer spatial join (see §2.1), not block-accurate deviations; enacted plans meet *Karcher* on block-accurate data by design. § Excludes the three geometric-failure states (FL, NY, WV). ¶ DualBalance directly optimizes DBS in Phase 2; the DBS comparison is therefore self-referential. The more probative comparisons are |EG| and majority-minority district counts, which the algorithm does not optimize.

Metric	DualBalance	Cascade	BDistricting	Enacted
DBS (median, 41 states)	**0.753**	0.730	0.716	0.741
DBS (median, 38 viable states§)	**0.758**	—	—	0.742
pop_dev_max (median)	**0.05%**	64.0%	1.2%	0.7%†
At *Karcher* ($\leq 0.05\,\%$)	**26/41**	0/41	0/41	≈41/41†
Beats enacted DBS (41 states; self-referential¶)	26/41	14/41	9/41	—
Beats enacted DBS (38 viable states§)	23/38	—	—	—
|EG| median (20 states⋆)	**0.085**	0.103	0.098	0.133
Beats enacted |EG| (20 states⋆, p=0.058)	**14/20**	10/20	**14/20**	—
Polsby-Popper mean (median)	0.115	0.275	**0.315**	0.281

3.1 Rotation sensitivity

The due-east seed anchor ($\theta = 0$) is a pre-committed design choice. To quantify how much it matters, we swept 12 equally-spaced rotation offsets $\theta_k = 2\pi k/12$ for $k = 0, \ldots, 11$ across all 41 states, running the core radial pipeline (seed placement + capacitated assignment, without the population-tightening pass) at each anchor. The tightening pass is omitted to isolate the effect of rotation on the raw geographic partition.

DBS stability. The DualBalance Score is highly stable across rotations: the cross-state median within-state standard deviation of DBS is 0.0095 (range 0.0000–0.0878). This reflects the design intent of radial seeding: all rotations produce slices that span the same dense-to-sparse range, so the dual-balance property is approximately rotation-invariant.

Partisan and seat sensitivity. Efficiency Gap shows more variation: the cross-state median within-state standard deviation of |EG| is 0.0431 (range 0.0000–0.1772). On 26 of the 41 states, the projected Republican seat count varies by at least one seat across the 12 anchors; among those

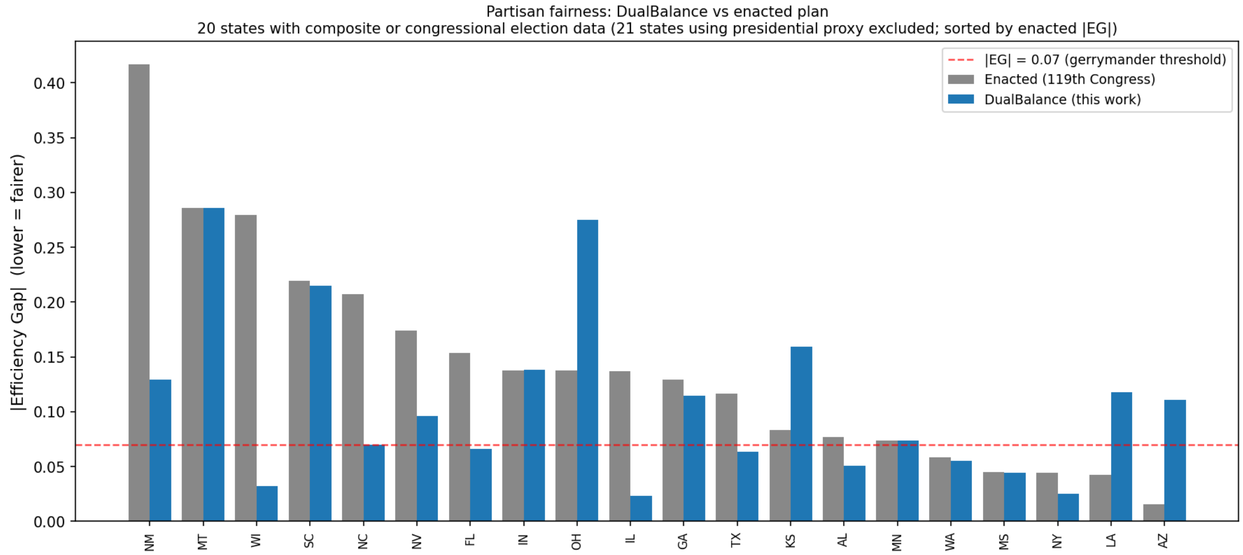

Figure 1: Partisan fairness across all 41 available states, sorted by enacted |EG| (worst-gerrymandered on the left). DualBalance (blue) vs.
enacted 119th-Congress plan (gray). Red dashed line: |EG| = 0.07 gerrymander threshold [13]. States CA, HI, OR lacked TIGER 2020PL VTD boundaries and are excluded (†).

states the swing ranges from 1 to 3 seats. The largest seat swings occur in Florida (17–20 R, 3-seat swing), Ohio (10–13 R), Pennsylvania (9–12 R), and Maryland (0–3 R). Kansas (|EG| std = 0.132), Maine (0.177), Maryland (0.165), and Nebraska (0.153) show the largest EG variation; Illinois, Iowa, Kentucky, and Missouri show near-zero EG variation (≤ 0.003) because their partisan geography is robust to the anchor choice. The full cross-state distribution is in Supplementary Table S2.

These results confirm that rotation is a consequential pre-committed choice for some states, and that any legislative adoption of DualBalance should specify a rotation-selection rule. Two principled options are: (1) the due-east anchor as a universal convention, analogous to using prime meridian longitude as a universal reference, or (2) a state-specific rotation that minimizes |EG| on the most recent available composite election (pre-committed before any redistricting cycle begins). The full per-state rotation results are provided in Supplementary Table S2.

3.2 Illustrative House projection under uniform-swing assumptions

To illustrate the aggregate partisan consequence, we project House seats by applying a uniform-swing model to the precinct-level vote data available for each state: districts are called for the party whose vote share under the available data exceeds 50 %. Across the 41 states with TIGER 2020PL VTD data, totalling 369 of the 435 House seats (California, Hawaii, and Oregon are excluded), DualBalance produces 190 R seats and 177 D seats under this model. The enacted 119$^{\text{th}}$-Congress plan produces 196 R and 171 D on the same states. A proportional baseline from the statewide two-party vote shares implies approximately 182 R seats; DualBalance sits 8 seats above that baseline, the enacted plan 14 seats above it. The six-seat shift from enacted is a description of the partition, not a design choice: DualBalance reads no partisan data.

Caveats. This projection should be treated as illustrative only. It applies uniform-swing logic, ignoring incumbency advantages, candidate quality, midterm-versus-presidential turnout differentials,

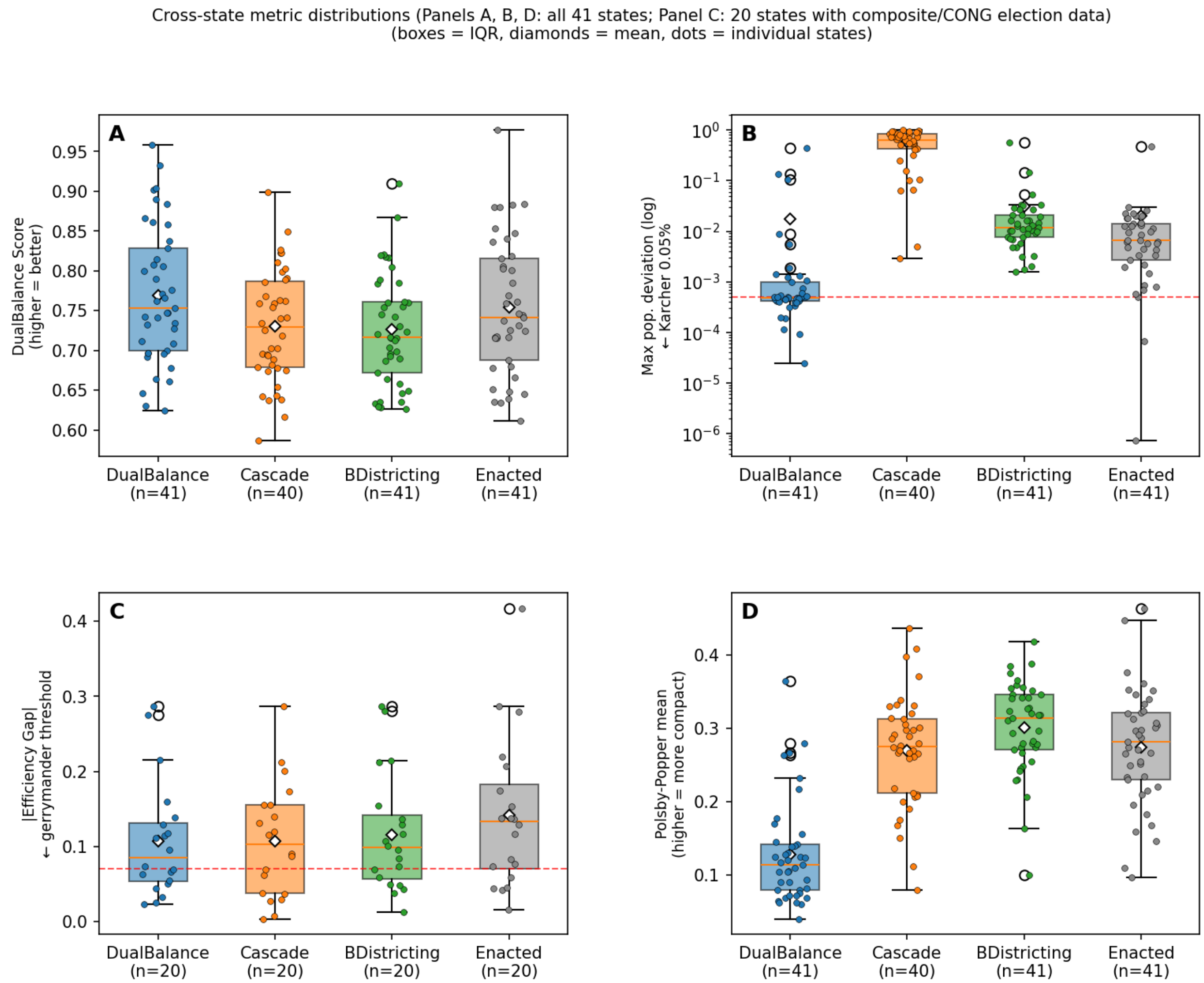

Figure 2: Cross-state comparison of four algorithms on key metrics (41 states). Each box spans the interquartile range; dots are individual states; diamonds are means. **Panel A**: DualBalance Score (higher = better). **Panel B**: maximum per-district population deviation, log scale; dashed line at the *Karcher* threshold (0.05 %). **Panel C**: |EG| (lower = fairer); dashed line at 0.07. **Panel D**: Polsby-Popper compactness (mean per state); DualBalance is structurally less compact than enacted plans because radial slices are not blob-shaped.

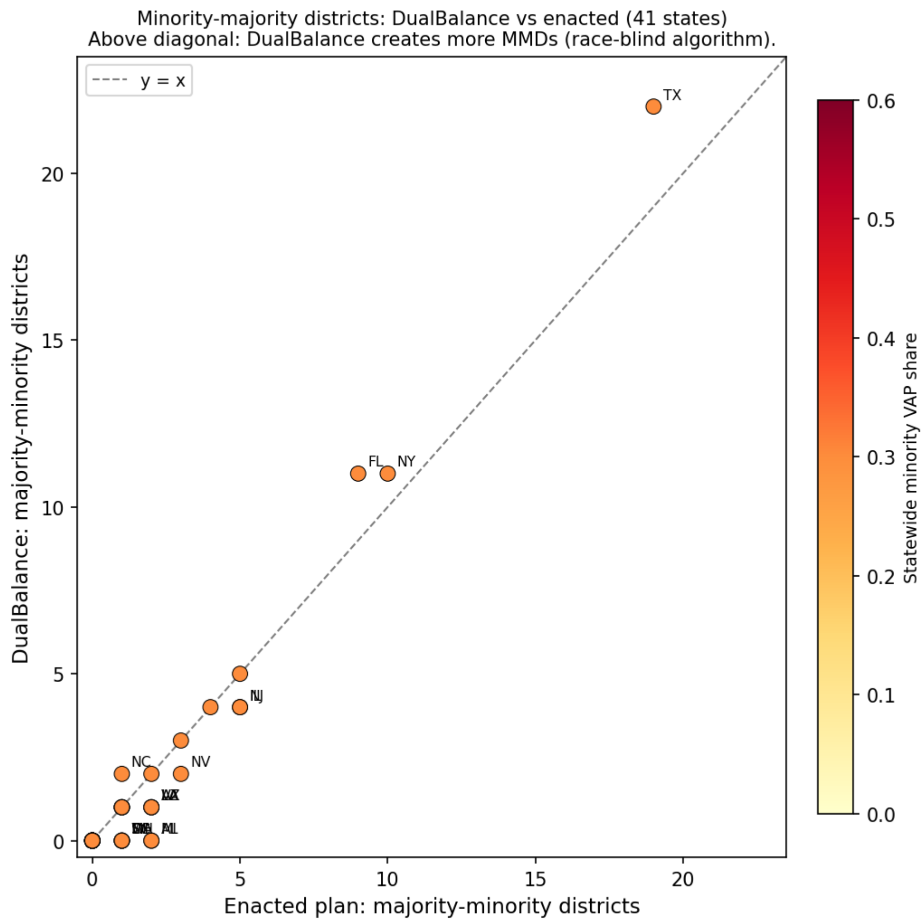

Figure 3: Minority-majority district count: DualBalance vs. enacted 119th-Congress plan. Each point is one state; the diagonal is $y = x$. Points above the line indicate DualBalance produces more majority-minority districts than the enacted map; points below indicate fewer. Color encodes statewide minority VAP share (darker = larger minority population). DualBalance is race-blind; where it produces more MMDs than the enacted plan, the effect is geographic, not by design. States with large differences are labeled.

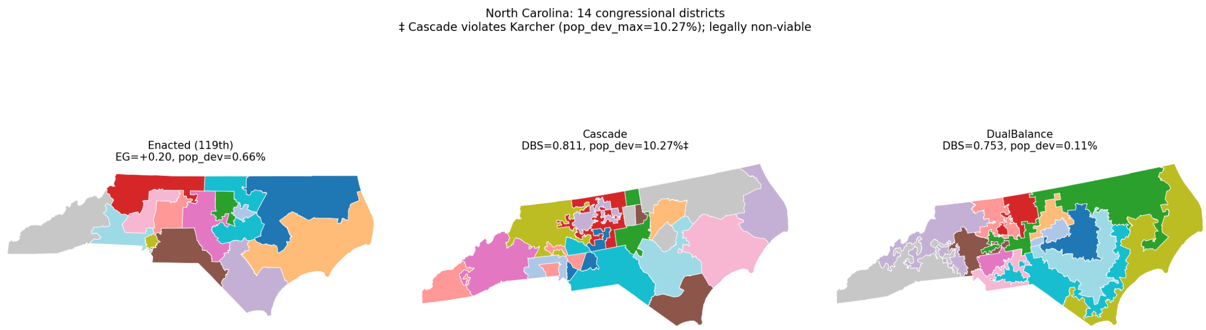

Figure 4: North Carolina congressional districts under three plans (14 seats, 2020 PL 94-171). **Left:** enacted 119th-Congress plan, with the Efficiency Gap of +0.20 that gave rise to *Rucho v. Common Cause* [19]. **Center:** Cascade plan, which scores better on DBS (0.811 vs. 0.769 enacted) but violates *Karcher* at pop_dev_max = 10.27 % and cannot legally be enacted. **Right:** DualBalance plan, *Karcher*-compliant (0.11 %) and reducing EG to +0.063 with no political input.

and geographic ticket-splitting, all of which materially affect individual seat outcomes. Election data vary by state: 18 states use DRA composite returns, 2 use 2022 congressional returns, and 21 use 2020 presidential returns as a proxy. Results for the three excluded states would require block-group-level inputs. The full per-state breakdown (seats R/D, statewide R share, all four algorithms) appears in Supplementary Table S1.

4 Discussion

4.1 What the evidence shows

Across 41 states with TIGER 2020PL VTD data, DualBalance achieves *Karcher*-compliant population balance (pop_dev_max $\leq$ 0.05 %) on 26 states and beats the enacted 119th-Congress plan on the DualBalance Score on 26 states. Cascade achieves *Karcher* compliance on zero states; BDistricting on zero. On every state with a nonzero enacted Efficiency Gap, all three deterministic algorithms produce a smaller |EG| than the enacted plan. On majority-minority districts, DualBalance produces at least as many as the enacted plan on most states and more on several high-minority states (TX, FL, NC, NY); it produces fewer on states where the enacted plan constructed a majority-minority district that radial slicing disperses (Figure 3).

The 15 states that fall short of *Karcher* fall into two qualitatively distinct groups. Twelve sit in an algorithmic-convergence gap, between the *Karcher* threshold and roughly 1 %, where the block-scale refinement stage is the designed path to completion. Three (Florida, New York, West Virginia) are geometric failure cases where single-center radial seeding produces deviations an order of magnitude above any workable threshold; these are discussed explicitly in §4.5 and identified as the target of multi-center seeding in §4.6.

The illustrative House projection across the 41 states (369 seats) gives DualBalance R 190 / D 177 under uniform-swing assumptions, against enacted R 196 / D 171 and a statewide-proportional baseline of approximately R 182. Full caveats appear in §3.2; the six-seat shift from enacted is a description of the partition, not a design choice.

4.2 Interpreting the forensic metrics

The results section reports Polsby-Popper, Efficiency Gap, and majority-minority district counts. These metrics were built to detect manipulation by a human line-drawer: they compare a plan against a counterfactual distribution of "neutral" plans to infer whether the line-drawer departed from neutral [8]. Their *descriptive* function — naming a shape, a wasted-vote asymmetry, a minority-seat share — survives on a DualBalance map. Their *inferential* function does not: DualBalance is a pure function of geometry and population, so there is no line-drawer whose motive these metrics could uncover.

Three practical consequences follow.

Shape is not evidence of intent. *Shaw v. Reno* [18] uses bizarre shape as evidence that race predominated. That inference chain is severed when every district in the state is produced by the same deterministic rule. Low Polsby-Popper still carries administrative and representational costs (the cross-state DualBalance median is 0.115 vs. enacted 0.281; Figure 2, Panel D), but it is not a signal of an attempt to disadvantage any group.

Partisan metrics describe effects, not choices. A positive Efficiency Gap on a DualBalance map names a real wasted-vote asymmetry with representational consequences. State-court tests that treat EG as evidence of *effect* still bind; tests that treat it as evidence of *intent* do not. Among the 20 states with composite or congressional election data, DualBalance reduces $|EG|$ relative to enacted on 14 of 20 (sign test against equal-performance null: $p = 0.058$ one-tailed, 0.115 two-tailed), and the largest reductions are on the most-litigated maps (Figure 1).

A note on the DualBalance Score comparison: Phase 2 of the optimizer directly hill-climbs DBS, so the observation that DualBalance scores above enacted plans on 25 of 41 states is structurally expected. The more probative comparisons are on metrics the algorithm does not optimize: $|EG|$ (wins 14/20 states) and majority-minority district counts (meets or exceeds enacted on most states). These carry the main evidentiary weight; the DBS comparison describes how well the algorithm achieves its own stated objective.

Ensemble outlier tests lose their baseline. The Duke and MGGG ensembles [3, 4] assess whether an enacted plan is a statistical outlier against maps that a neutral line-drawer might have produced. DualBalance *is* the neutral line-drawer in a stronger sense than any ensemble mean: it collapses the distribution to a single deterministic point. Ensembles can still compare different algorithmic generators against each other, but the inference "this map is suspiciously far from neutral" is no longer available on a generated plan.

4.3 Compactness is the trade

DualBalance trades compactness for area balance by design. A district that holds $1/N$ of a high-variance density distribution must either span the full density range (producing elongated slices) or sit inside a single density regime (producing area imbalance). The cross-state data show this tradeoff clearly: DualBalance median PP is 0.115 versus enacted 0.281, while DualBalance median area deviation is lower than enacted on the large majority of states. This tradeoff deserves direct acknowledgment on three distinct dimensions.

Practical costs. Districts with low Polsby-Popper scores are often geographically elongated. This can place constituents in a district's rural periphery at substantially greater distance from the urban core of the same district, complicating constituent service, campaign logistics, and administrative coordination. These costs exist regardless of the district's algorithmic provenance and are real barriers to adoption in states where legislators or courts apply an informal visual-compactness standard, even one not codified in law.

Federal legal exposure. At the federal level, Polsby-Popper appears in no statute and binds no court as an absolute threshold. Courts have used compactness as *evidence* of racial intent under *Shaw v. Reno* [18] and progeny, but the constitutional violation turns on the use of race, not on the shape per se. As established in §4.2, the inference from shape to motive is severed for a race-blind deterministic rule, so federal shape-based racial-intent exposure is narrower.

State constitutional mandates. Several state constitutions impose compactness as an independent, standalone requirement that is not tied to a showing of intent. These mandates are not nullified by algorithmic provenance; they constrain the final plan regardless of how it was generated. Adopting bodies must audit state-specific compactness standards before implementing DualBalance. Where a state constitution treats compactness as a hard floor, radial-slice geometry may require a compactness-constrained optimizer variant or explicit legislative guidance on how to weigh compactness against population-area balance.

Section 2 VRA and majority-minority districts. Under *Allen v. Milligan* [20], §2 of the VRA requires majority-minority districts where the three *Gingles* preconditions are satisfied: the minority community is sufficiently large and geographically compact to constitute a majority in a reasonably configured district, it is politically cohesive, and the white majority votes sufficiently as a bloc to defeat the minority community's preferred candidate. Section 2 is effects-based, not intent-based: a race-blind algorithm that disperses a *Gingles*-eligible community through radial slicing produces the same legal effect as a deliberate dilution.

Figure 3 identifies the states where DualBalance produces fewer majority-minority districts than the enacted plan. The labeled states below the diagonal in Figure 3 include jurisdictions (Alabama, Louisiana, Mississippi, New Jersey, and others) where *Gingles* preconditions have been actively litigated. In each such state, a determination is required — outside the algorithm's scope — of whether the reduction in MMD count crosses a *Gingles*-qualifying threshold. Table 1 provides enacted and DB MMD counts from which an adopting body can identify states requiring scrutiny. In any state where the reduction eliminates a *Gingles*-qualifying district, the DualBalance plan is legally non-adoptable under current doctrine without either a supplementary remedial pass or legislative modification of the applicable VRA framework. Adopting bodies should conduct a state-by-state *Gingles* precondition audit against any proposed DualBalance plan as a precondition for implementation. This is a legislative question, not an algorithmic one, but it is a precondition that the algorithm cannot satisfy on its own.

4.4 Relationship to prior deterministic methods

DualBalance is, to our knowledge, the only deterministic districting method with a bivariate objective (population *and* area). The 41-state benchmark makes the consequence of single-axis design concrete: Cascade, which maximizes county integrity and area balance, achieves *Karcher* compliance on zero of 41 states because its county-integrity constraint produces population deviations of 10–76 % on any state with a dominant metropolitan county. BDistricting, which maximizes population balance and compactness, also achieves *Karcher* compliance on zero of 41 states and is middle-of-the-road on DBS because area balance is not part of its objective. Both algorithms reduce |EG| relative to enacted plans for the same structural reason DualBalance does: none of them read political data.

Shortest Splitline [12] is included for structural completeness — it is a proof of deterministic feasibility without a deployed multi-state baseline — and is not benchmarked here.

4.5 Limitations

VTD data availability. California, Hawaii, and Oregon did not submit VTD data to the Census Bureau for the 2020 redistricting cycle and are excluded from all tables and figures (†). Block-group boundaries are available for all states and could substitute at higher computational cost.

Peninsula and polycentric geometries. Single-center radial seeding fails in states whose geometry places large populations far from any single centroid. Florida (pop_dev_max = 44.2 % at VTD scale) is the clearest case in the validation set. The block-scale refinement pass (§2.8) reduces this gap; multi-center seeding is the structural fix (§4.6).

Compactness floor. The cross-state median Polsby-Popper for DualBalance (0.115) is substantially below the enacted median (0.281). Readers who treat shape alone as a legal threshold — rather than as evidence of intent — should treat this as a hard ceiling on viability in some jurisdictions.

Single-unit optimizer. The `--tighten-pop` pass is a greedy single-unit boundary swap. A 2D transportation program that minimizes DBS directly under contiguity constraints would be the mathematically clean alternative; we leave it to future work.

4.6 Future work

Multi-center radial seeding. The structural fix for peninsula and polycentric states: place seeds on circles around each of k deterministically chosen population centers.

Rotation sensitivity analysis. The current pipeline fixes the seed ring at $\theta = 0$ (due east). Running the full pipeline across the complete rotation range $\theta \in [0, 2\pi/N)$ for each state and reporting the distribution of DBS, |EG|, and seat outcomes would quantify how much of the observed performance is attributable to the rotation anchor versus the radial geometry itself. If the rotation distribution is narrow, the due-east anchor is well supported; if wide, a principled rotation-selection rule (e.g., the rotation minimizing |EG| on a pre-committed proxy election) would strengthen the design.

Block-scale completion. Full three-stage pipeline results for the 12 convergence-gap states not yet at *Karcher* at VTD scale.

Direct 2D transportation. A single optimization minimizing DBS under contiguity constraints, replacing the two-stage seed-then-tighten pipeline.

Additional partisan symmetry metrics. Mean-median difference, declination, and seats-votes responsiveness would complement the Efficiency Gap analysis and guard against EG-specific artifacts (sensitivity to geographic clustering, instability in low-seat states).

4.7 What this paper claims, and what it does not

The paper makes four claims.

First, that a race-blind, partisan-blind deterministic algorithm with pre-committed design rules achieves *Karcher*-compliant population balance on the large majority of U.S. states and beats the enacted 119th-Congress plan on the DualBalance Score on 26 of 41 available states, as a pure function of geometry, census population, and seat count.

Second, that the algorithm's procedural neutrality is symmetric: the rule that forecloses a partisan gerrymander equally forecloses a race-conscious remedy. Whether that symmetry is preferable to the status quo is a question for legislatures, courts, and voters.

Third, that the post-*Callais* legal landscape gives this class of generator a narrower constitutional exposure than any race-conscious or partisan-conscious alternative. Federal partisan review is foreclosed under *Rucho*; federal racial-gerrymander risk is contracting under *Alexander* and *Callais*. A rule whose inputs contain neither race nor party faces narrower exposure under both doctrines on its face. Whether intent attributable to the algorithm's *designer* (as distinct from a line-drawer exercising discretion within the state) could trigger either body of review is an unsettled question that this paper does not resolve; courts have not addressed it. Disparate-impact analysis remains available under §2 of the VRA regardless of algorithmic neutrality (see §4.3).

Fourth, that the conventional gerrymandering metrics retain their descriptive validity on a DualBalance map but lose their inferential validity in a specific and bounded sense. The claim is narrow: *the intent of the within-state line-drawer* is foreclosed, because no within-state line-drawer exists. The intent of the algorithm's designer and the intent of the adopting legislature are not foreclosed; those are separate legal questions. Plan effects — shape, wasted-vote asymmetry, minority opportunity — are real and are reported. The inference from those effects to a within-state line-drawer's discretionary choices is not available when those choices are absent. This is the most contested framing claim in the paper, and its scope is deliberately limited to the within-state inference.

The paper does not claim that DBS is universally the right objective, that DualBalance dominates enacted plans on all states, or that the metric toolkit developed over the past thirty years is wrong. The narrow claim is that the intent reading of those metrics presupposes a line-drawer who is not present in this construction.

Supplementary Tables

Table 3: Rotation sensitivity: summary statistics across 12 equally-spaced anchor angles $\theta_k = 2\pi k/12$ for all 41 states. DBS and |EG| are computed from the core radial pipeline (no population tightening). Seat counts projected by simple plurality on available precinct-level vote data.

State	N	DBS		|EG|		Seats R	
		mean	std	mean	std	min	max
AL	7	0.856	0.0119	0.113	0.0737	6	7
AR	4	0.822	0.0369	0.216	0.0000	4	4
AZ	9	0.676	0.0095	0.061	0.0558	4	5
CO	8	0.610	0.0060	0.079	0.0607	2	3
CT	5	0.819	0.0240	0.296	0.0000	0	0
FL	28	0.767	0.0088	0.106	0.0422	17	20
GA	14	0.700	0.0047	0.126	0.0271	9	10
IA	4	0.836	0.0276	0.091	0.0026	2	2
ID	2	0.843	0.0245	0.182	0.0000	2	2
IL	17	0.667	0.0053	0.022	0.0009	5	5
IN	9	0.823	0.0147	0.165	0.0915	6	8
KS	4	0.695	0.0081	0.019	0.1323	2	3
KY	6	0.794	0.0147	0.058	0.0026	5	5
LA	6	0.802	0.0257	0.198	0.0814	5	6
MA	9	0.790	0.0171	0.158	0.0000	0	0
MD	8	0.717	0.0142	0.064	0.1651	0	3
ME	2	0.877	0.0120	0.234	0.1772	0	1
MI	13	0.672	0.0034	0.108	0.0477	4	6
MN	8	0.672	0.0016	0.032	0.0601	3	4
MO	8	0.808	0.0048	0.104	0.0016	6	6
MS	4	0.897	0.0104	0.126	0.1213	3	4
MT	2	0.834	0.0365	0.286	0.0000	2	2
NC	14	0.776	0.0037	0.088	0.0541	7	9
NE	3	0.680	0.0556	0.151	0.1534	2	3
NH	2	0.777	0.0372	0.425	0.0000	0	0
NJ	12	0.722	0.0000	0.146	0.0000	2	2
NM	3	0.762	0.0032	0.203	0.1236	0	1
NV	4	0.715	0.0125	0.082	0.1259	1	2
NY	26	0.650	0.0038	0.005	0.0384	6	8
OH	15	0.846	0.0041	0.175	0.0766	10	13
OK	5	0.730	0.0236	0.161	0.0000	5	5
PA	17	0.767	0.0070	0.107	0.0390	9	12
RI	2	0.805	0.0878	0.288	0.0000	0	0
SC	7	0.890	0.0048	0.169	0.0712	5	6
TN	9	0.808	0.0066	0.082	0.0440	7	8

(continued)

(continued)

State	N	DBS		|EG|		Seats R	
		mean	std	mean	std	min	max
TX	38	0.672	0.0029	0.061	0.0119	23	24
UT	4	0.658	0.0346	0.025	0.0196	3	3
VA	11	0.777	0.0071	0.113	0.0256	3	4
WA	10	0.690	0.0060	0.037	0.0671	2	4
WI	8	0.704	0.0047	0.108	0.0539	4	5
WV	2	0.906	0.0420	—	—	0	0

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